Massive Bioinformatics - "Mini-Bioinformatics Pipeline" Case Study Report

Date: March 11, 2026 Arda Baran Kılınç

Project Summary and Source Codes:

Within the scope of this project, a reproducible Quality Control (QC) workflow was designed for long-read sequencing data.

- The project environment was isolated using Conda with the environment.yml file.
- Workflow (pipeline) automation was achieved using Nextflow.
- For QC analyses, both the industry-standard NanoPlot tool and a custom Python script for data processing and visualization were utilized.

All source codes, configuration files, and complete result outputs for the project can be accessed from the GitHub repository below:

GitHub Repository: https://github.com/ardaabk/qc-pipeline-for-internship

Status Assessment Email for Professor Kılıç

Subject: Quality Control (QC) Results for Long-Read Sequencing Data

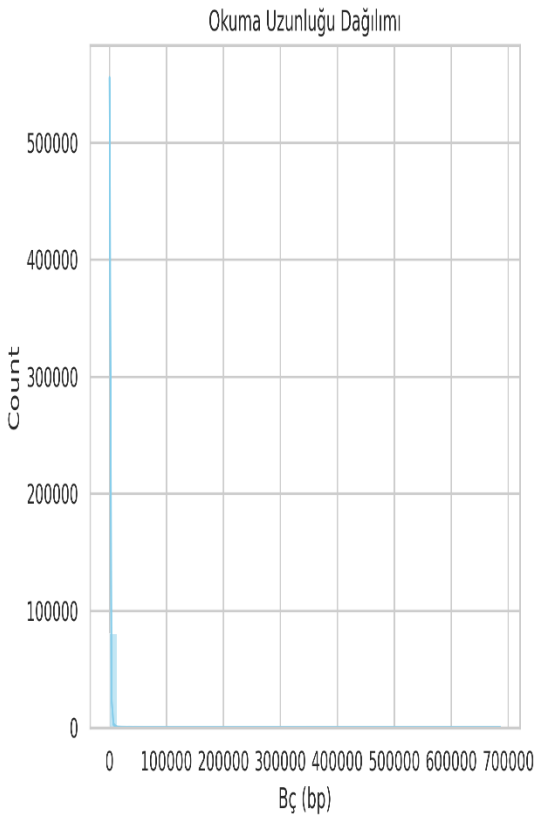
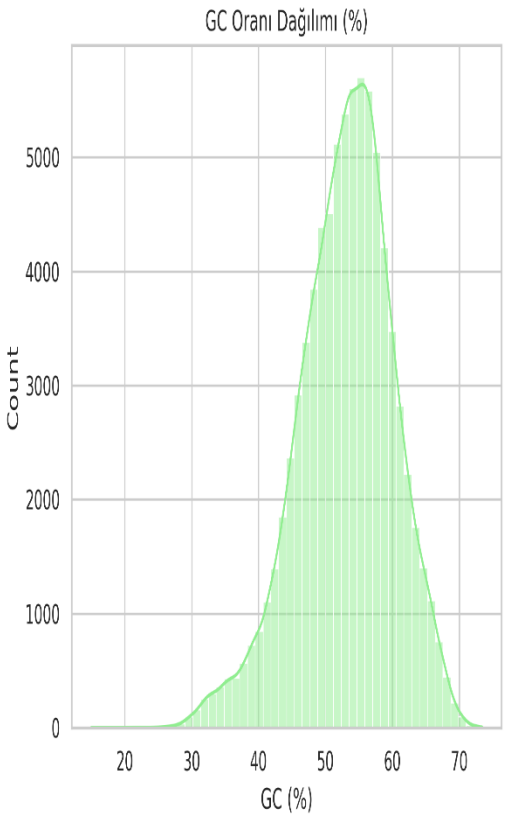
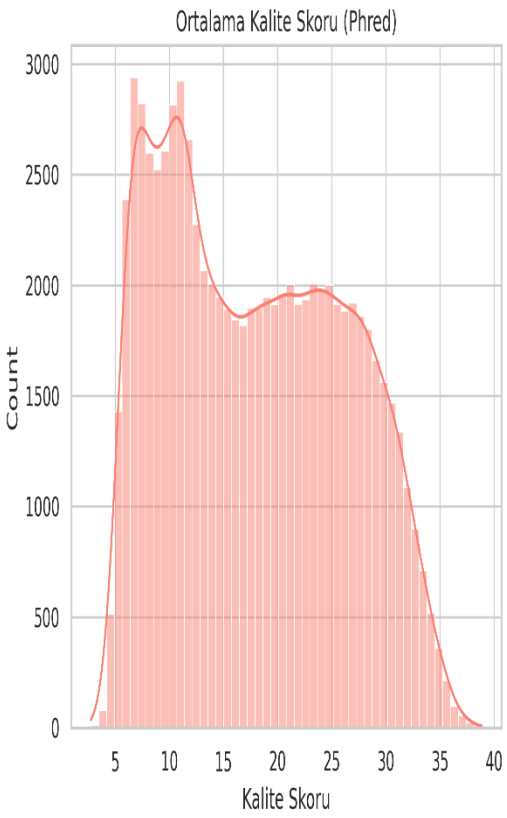
Dear Professor Kılıç,

I have completed the necessary quality control (QC) analyses on the barcode77.fastq sequencing data provided by your laboratory. Below is a summary of the results obtained using my custom Python script and the NanoPlot tool:

- **Read Lengths:** The vast majority of the reads in the dataset are concentrated in the 200 - 400 base pair (bp) range. This profile is consistent with our targeted lengths.
- **Mean Quality Score:** The Phred quality scores of the reads predominantly range between Q20 and Q33. For the Nanopore platform, these Q20 and above values indicate that the sequencing was performed with over 99% accuracy, confirming that the data is of exceptionally high quality.
- **GC Content:** The distribution graphs demonstrate that the GC ratio exhibits a balanced, unimodal distribution, generally in the 45%-55% range. This verifies that there is no significant external contamination in the sequenced sample.

The dataset easily exceeds our quality standards. We can confidently proceed to the next steps (assembly or alignment) without anticipating any issues.

Best regards,

Arda Baran Kılınç Bioinformatics Intern Candidate

APPENDIX: Analysis Visuals